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Received: 15 September 2023 | Revised: 9 November 2023 | Accepted: 3 January 2024

DOI: 10.1111/1556-4029.15469

JOURNAL OF FORENSIC SCIENCES AAFS

ORIGINAL PAPER

Criminalistics; Pathology/Biology

Investigative genetic genealogy for human remains identification

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Abstract

Investigative genetic genealogy (IGG) has emerged as a highly effective tool for tying a forensic DNA sample to an identity. While much of the attention paid to IGG has focused on cases where the DNA is from an unknown suspect, IGG has also been used to help close hundreds of unidentified human remains (UHR) cases. Genome-wide single-nucleotide polymorphism (SNP) genotype data can be obtained from forensic samples using microarray genotyping or whole-genome sequencing (WGS) with protocols optimized for degraded DNA. After bioinformatic processing, the SNP data can be uploaded to public GG databases that allow law enforcement usage, where it can be compared with other users' data to find distant relatives. A genetic genealogist can then build the family trees of the relatives to narrow down the identity of the source of the forensic DNA sample. To date, 367 UHR identifications using IGG have been publicly announced. The same IGG techniques developed and refined for UHR cases have significant potential for disaster victim identification, where DNA is often extremely compromised, and close family references may not be available. This paper reviews the laboratory, bioinformatic, and genealogical techniques used in IGG for UHR cases and presents three case studies that demonstrate how IGG is assisting with remains identification.

KEYWORDS
disaster victim identification, forensic genetic genealogy, investigative genetic genealogy, single-nucleotide polymorphism, unidentified human remains, whole-genome sequencing

Highlights

- Investigative genetic genealogy (IGG) has helped identify 100s of unidentified human remains (UHRs).
- Specialized laboratory and bioinformatic methods have been developed for challenging UHR samples.
- Techniques developed and refined for UHR cases can also be used to identify disaster victims.
- UHR identity can be confirmed using SNP kinship testing against a voluntary close family reference.
- To date, 367 UHR identifications using IGG have been publicly announced.

[Correction added on February 14, 2024 after first online publication: The number of UHR identifications using IGG has been revised in this version.]

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1 | INTRODUCTION

Investigative genetic genealogy (IGG) has emerged as a highly effective tool for tying a forensic DNA sample to an identity [1]. While much of the attention paid to IGG has focused on cases where the DNA is from an unknown suspect, IGG has also been used to help close hundreds of unidentified human remains (UHR) cases. This paper discusses the proven DNA processing and bioinformatic techniques for obtaining high-quality single-nucleotide polymorphism (SNP) data from forensic samples and the genetic genealogy methods used to narrow down possible identities from UHR cases. These same specialized techniques can also be applied to disaster victim identification (DVI), where DNA samples often present the same challenges seen in UHR cases with low-quality samples [2]. When family references are not available or when family references are too distant for traditional kinship testing using STRs or mitochondrial DNA, IGG can assist with DNA-based identification. To date, 367 identifications using genetic genealogy in UHR cases have been publicly announced, all in the United States Canada [3]. Many of these UHR identifications have been in cases that are decades old, but utilizing IGG for more recent cases has become increasingly common, with 17.7% of publicly reported IGG identifications occurring within 5 years of the discovery of the UHR (Figure 1). For example, in 2018, IGG was used to help identify Camp Fire victims [4], which is, to the authors' knowledge, the only time it has been used in a DVI scenario thus far. This paper reviews the techniques involved in IGG that have been developed and refined for UHR, which could be applied to DVI cases, and presents three demonstrative UHR case studies.

2 | METHODS

2.1 | Methods for generating and processing genome-wide SNPs

Unlike traditional forensic DNA analysis, IGG requires data from hundreds of thousands of SNPs across the genome. High-quality SNP data at that scale from bone, teeth, and tissue samples can be obtained using two different laboratory techniques: microarray genotyping or whole-genome sequencing (WGS). For Parabon's UHR cases resulting in an identification, 50.6% have used microarray and 49.4% have used WGS. Microarray chips target a specific set of SNPs using short 50-bp probes; for example, the CytoSNP-850K chip targets ~850,000 SNPs. Each SNP has two probes, one for each allele, that fluoresce in different colors (red or green) when DNA hybridizes with them. A scanner reads the total amount and ratio of the fluorescence, and then software compares those values to predefined clusters for each genotype. Samples that fall within a cluster are called as having that genotype. The overall quality of the microarray results can be evaluated as the call rate, the proportion of targeted SNPs with reported genotypes. As part of our application for approval from the New York Department of Health (NYDOH), the CytoSNP-850K chip and Parabon's forensic microarray protocol were validated down to 1 ng of DNA input for most forensic sample types. Microarray genotyping on the Illumina Global Screening Array (GSA) using the standard protocol has been validated down to 0.2 ng for intact samples [5]. However, SNP recovery on the GSA has been lower on degraded samples [6]. It is currently not known why there has been a greater level of genotyping success for degraded samples on the CytoSNP-850K chip with Parabon's forensic protocol compared with the GSA using the standard protocol.

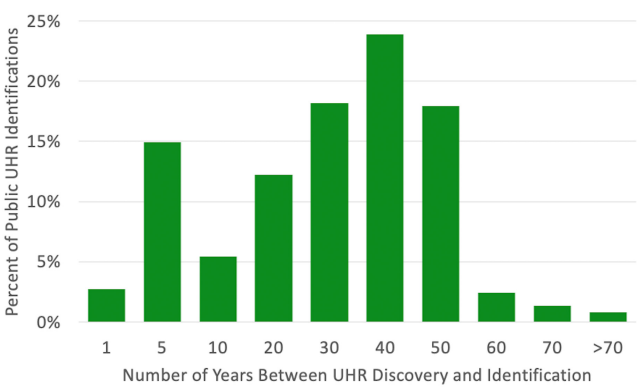


FIGURE 1 Number of years between unidentified human remains (UHR) discovery and identification for publicly announced UHR IGG identifications [3].

Though microarray uses short probes, it has proven less effective for highly degraded samples that have a high proportion of non-human DNA, such as bone samples [7]. Therefore, we use WGS for most bone samples, samples with <1 ng of DNA (down to 114 pg thus far), and highly degraded samples. When QC indicates the proportion of human DNA is low, we also use the Arbor myBaits human whole-genome enrichment (WGE) kit [7] to enrich for the human DNA portion of the sample. The importance of WGE can be visualized as the difference in alignment rate to the human genome between the pre-enrichment and post-enrichment samples (Figure 2). WGS on bone and Parabon's bioinformatics have been validated down to 50 pg of DNA input.

While microarray analysis produces genotype calls, WGS produces sequencing reads that require additional bioinformatic processing to generate genotype calls. The raw sequencing reads must first be mapped (aligned) to the human reference genome to determine the location they originated from. This process also removes sequencing reads that originated from non-human sources (e.g., bacterial and fungal species related to decomposition). Genotypes for SNPs of interest can then be determined by looking at the number of reads that match the reference genome and the number that differ from the reference genome. Genotype calculations also take into account the quality of the sequence data and the possibility of sequencing errors. More sequencing reads covering each location result in more confident genotype calls. Studies with high-quality DNA typically aim to have an average of 30 reads covering each location ("30X coverage"), but when sequencing low-quality samples, this is often not a realistic goal. Less than 30% of our WGS samples have over 5X coverage (Figure 3).

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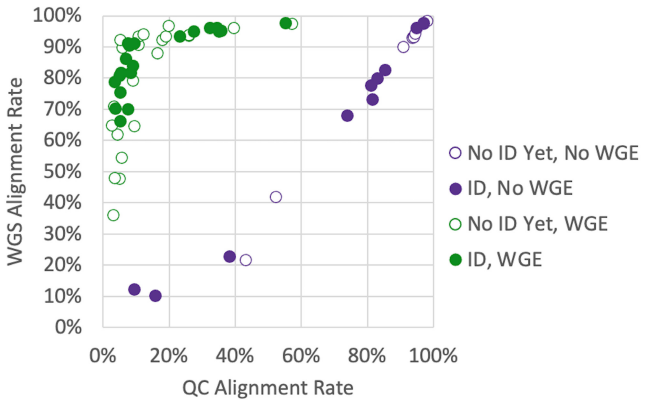


FIGURE 2 Alignment rate to the human genome for a selection of Parabon casework samples from 30 solved and 30 unsolved cases at the pre-enrichment (QC) and post-enrichment (WGS) stages for samples with (green) and without (purple) whole-genome enrichment (WGE). Cases that have resulted in an identification (ID) are shown as filled points and cases that have not yet resulted in an identification to date (No ID) are shown as open points. Plot only includes samples selected from the subset of cases for which QC sequencing, full sequencing, and IGG were performed.

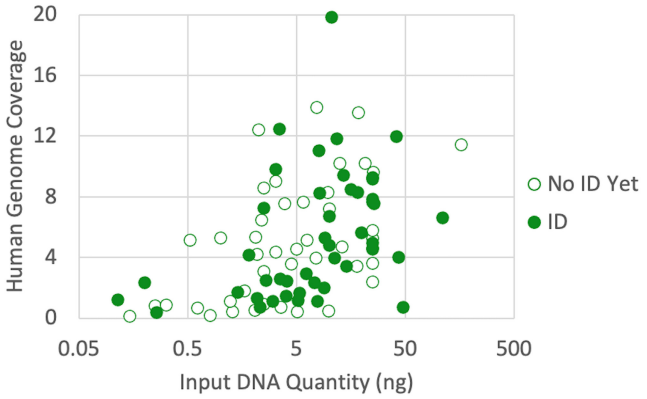


FIGURE 3 Human genome coverage versus input DNA quantity in ng for a selection of 45 solved and 45 unsolved Parabon WGS cases where IGG was performed. The smallest quantity of DNA used for a Parabon IGG case is 114 pg, and that case has been solved. The lowest coverage that has resulted in an identification is 0.39X. Cases that have resulted in an identification (ID) are shown as filled points and cases that have not yet resulted in an identification (No ID) are shown as open points.

Low coverage means that many SNPs of interest do not have any sequencing reads or not enough reads to make a confident genotype call. However, because the whole genome has been sequenced and not just the SNPs of interest, we can use sequence reads that map to other "tag" SNPs that are linked to our "target" SNPs, meaning they are not inherited independently, to call genotypes using a process called low-coverage imputation (LCI) [8]. LCI is distinct from traditional imputation in that, rather than imputing many rare variants from a few common variants (traditional imputation), LCI imputes a few common variants (microarray SNPs) from many rare variants (WGS data). LCI is expected to produce highly accurate genotypes for common SNPs (minor allele frequency > 0.05) down to very low coverage (below 1X). First, genotype likelihood scores are generated for a set of tag and target SNPs, which are then used as inputs to GLIMPSE [9] to calculate genotype probabilities using a reference panel of haplotypes from the 1000 Genomes Project. The results are filtered to high-probability genotype calls and used as inputs to Beagle 5.1 [10] for imputation, after which the results are again filtered to target SNPs with high-probability genotype calls. Using simulated low-coverage sequencing data from the 1000 Genomes, Parabon has validated this analysis pipeline down to below 0.5X coverage [11], and it has been successfully used for identification via IGG on casework samples down to 0.39X coverage [12].

In cases with extremely degraded or damaged remains, such as mass disasters, IGG using genome-wide SNPs may not be an option due to the condition of the DNA. Qiagen's KIntelligence assay uses PCR probes to target 10,230 SNPs for sequencing. Paired with a custom matching algorithm, it has been reported that the KIntelligence kit SNP set can detect out to fifth-degree relatives (second cousins) with 56% sensitivity and nearly 100% specificity [13]. The assay is reportedly robust to degradation and low DNA input amounts [14], and the matching method is robust to allele dropout. We have not yet used KIntelligence for Parabon cases, but in our experience, many IGG cases do not have a top match at the fifth-degree level (~200 cM) or closer, and more distant matches are usually needed to support the conclusions as well (see case studies for examples). More than 50% of Parabon's cases resulting in an identification had a top match below 200 cM (Figure 4), and this number is even higher within non-UHR cases, where the GEDmatch database is smaller. Thus, KIntelligence is likely to be most useful in UHR cases, where the probability of obtaining a closer match is higher, rather than in suspect cases.

In some UHR cases, there may be a limited pool of possible identities and IGG is not needed. Most kinship tools using autosomal SNPs require high-coverage sequencing data for direct genotype calling, which is often not possible with challenging samples, especially when there is a high proportion of non-human DNA. In scenarios where

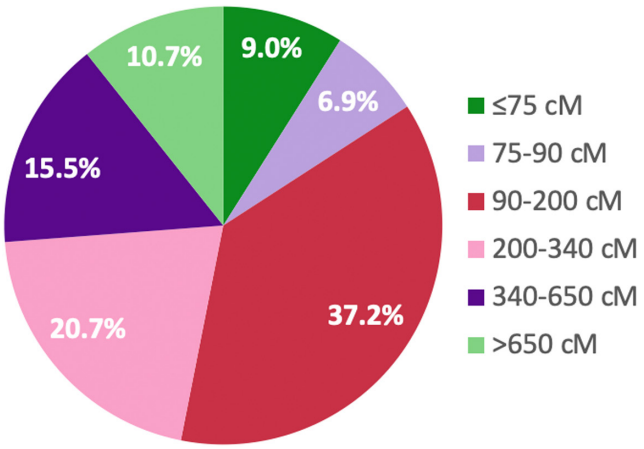


FIGURE 4 Amount of shared DNA (in cM) between the forensic sample and the top GG database match in Parabon's nearly 300 IGG cases that have resulted in an identification.

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no mitochondrial DNA reference individual is available, or the UHR has a common mitochondrial haplotype, autosomal kinship testing using low-coverage sequencing data offers an alternative approach. Rather than calling genotypes, low-coverage kinship calculates the likelihood of the observed data under various relationship hypotheses [15]. In 2022, the Armed Forces DNA Identification Laboratory (AFDIL) successfully used Parabon's low-coverage kinship software to connect non-probative World War II-era remains, some of which had been chemically treated with formaldehyde and powdered hardening compounds [16], to microarray genotyping data from family references out to fourth degree [17]. Shotgun sequencing of these samples yielded extremely poor data (90% of samples had <1000 SNPs at 1X coverage after damage correction). However, SNP capture of ~25,000 or ~95,000 SNPs was highly successful, yielding an average of 15,204 and 42,247 SNPs, respectively, with at least 1X coverage and enabling matching to the correct family. AFDIL is currently validating this kinship tool, as implemented in their Parabon Fx™ Forensic Analysis Software Platform deployment, for their SNP capture and WGS casework to identify fallen soldiers from past conflicts.

2.2 | Investigative genetic genealogy

2.2.1 | Uploading to IGG databases

Whether using microarray or WGS, the goal is to generate a file of genotypes that can be uploaded to genetic genealogy (GG) databases for distant relative searching. The total proportion of SNPs with called genotypes is known as the call rate, with a higher call rate generally indicating higher-quality data (Figure 5). Two databases are currently available to law enforcement (LE) for UHR identification: GEDmatch and FamilyTreeDNA (FTDNA). Each provides a specific mechanism for LE to upload data for a fee: GEDmatch PRO for GEDmatch and an approval process for FTDNA.

The GEDmatch and FTDNA databases contain genome-wide SNP profiles from individuals who have tested their DNA with a direct-to-consumer (DTC) company through microarray SNP genotyping, each of which is termed a "kit." Individuals who test with FTDNA are automatically entered into the FTDNA database, while users uploading to FTDNA or GEDmatch must download their data from a DTC company and upload it to the database. Both databases offer users the option to opt in or opt out of LE searches; however, unlike suspect cases, GEDmatch allows UHR cases to be searched against the entire public database. In both databases, LE kits do not appear on non-LE users' match lists. Individuals of Northern and Western European ancestry are heavily represented in these databases due to the demographics of consumers taking DTC genetic tests. For example, more than 70% of GEDmatch users are from the United States or Canada [18], and thus far, IGG has seen limited application outside North America. Thus, the majority of UHR cases that have resulted in an identification are of individuals with European ancestry [3] (Figure 6).

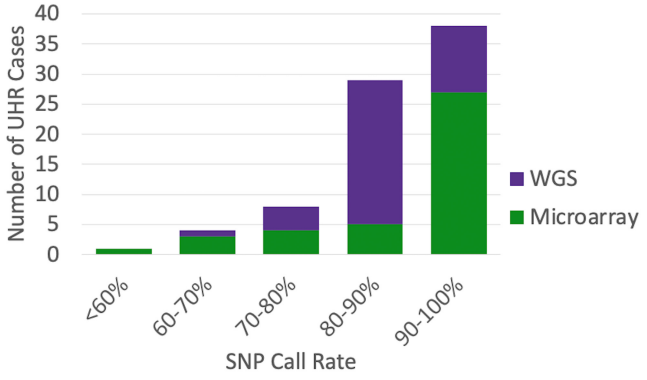


FIGURE 5 Call rates for Parabon's unidentified human remains (UHR) cases that have resulted in an identification using microarray (green) or WGS (purple).

2.2.2 | Ethics considerations for IGG

There have been several ethics concerns brought up regarding IGG, most of which only apply in IGG cases where the goal is to help identify a perpetrator. This section summarizes our points of view about those ethical considerations that are relevant to UHR cases, which generally relate to the genetic privacy rights of the individuals participating in GG databases. While a person's choice to have their DNA tested and to have their data in a database can put them at risk of having their genetic information unintentionally breached, LE's usage of a GG database does not increase this risk in any way, as no genetic information is accessed during IGG investigations. Users of GEDmatch and FTDNA may opt in or opt out of LE matching, and those who have not opted in are not seen by investigators, as long

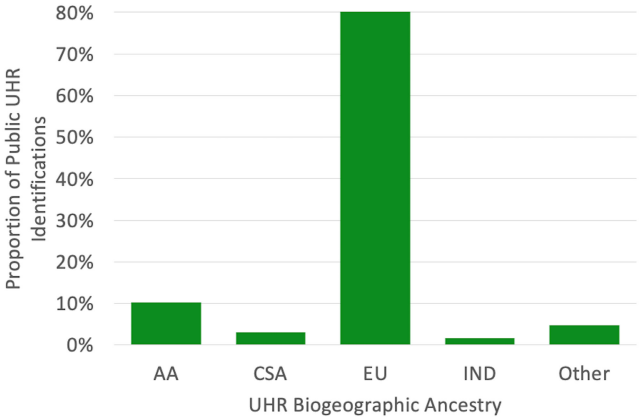


FIGURE 6 Reported biogeographic ancestry (BGA) of individuals identified in public UHR IGG cases [3]. Populations are labeled as in the original dataset: AA, African American Ancestry; CSA, Central/South American Ancestry, Including Mexico; EU, European BGA; IND, Indigenous Ancestry; Other includes East Asian Ancestry, Indian Subcontinent Ancestry, Middle Eastern and North African Ancestry, Sub-Saharan African Ancestry, and individuals with multiple ancestries.

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as all LE samples are correctly uploaded as LE profiles and the databases are functioning properly. However, GEDmatch allows investigators to compare UHR samples to the entire public database, so it is possible to opt out of LE matching on that site and still appear in comparisons with samples from LE cases. The alternatives are to not use the site or to set one's kit to "Research" or "Private."

2.2.3 | Analysis of database matches

Throughout this section, the source of the UHR DNA will be referred to as the unidentified Subject, or simply the Subject. FTDNA and GEDmatch compare each individual in the database to the Subject's kit and calculate the amount of DNA they share identical-by-descent (IBD) [18], returning a list of kits that share a significant amount of DNA, termed "matches." Briefly, IBD segments are long stretches of hundreds of SNPs over which the two individuals share at least one allele. Segments are measured in centimorgans (cM) where one cM equates to a 1% probability of recombination between the two genomic locations. Segments must be ≥7 cM to be counted. Sharing alleles across so many SNPs in a row over such a long genetic distance means it is much more likely that the segment was inherited from a common ancestor (i.e., that the two individuals are related) than that the similarity occurred randomly. The shared segments are then summed across the autosome to get the total amount of shared autosomal DNA (Figure 7). The amount of DNA shared on the X-chromosome is also shown. On both sites, the DNA data itself is not shown, only the lengths and locations of the shared segments, which can be visualized in a chromosome browser.

Using a tool such as DNA Painter [19], the amount of shared DNA can be converted to the probability that the two individuals share each degree of relationship (Figure 8). Typically, closer relatives share more DNA. This tool is based on thousands of self-reported data points of shared DNA between people with known relationships. The range of relationship probabilities can be translated to the range of the numbers of generations between the individuals and their common ancestor(s).

2.2.4 | Genealogy research

A genetic genealogist's first task upon receiving a GG database match list is to attempt to identify the matches. Some individuals have very common names, making it difficult to determine exactly which individual is in the database. Additionally, users have the option to use an alias, which, combined with an anonymous email

(A)

Autosomal			X-DNA						Haplogroup			
Total cM	Largest	Gen	Total cM	Largest	ICW Tool	Source	Overlap	Age(days)	Mt	Y	Name (* => alias)	Email
253.8	60.8 Q	2.91	0	0	Match	Ancestry	197577	1652	H11a		*E.D.F.	*****@aol.com
148.4	48 Q	3.30	0	0	Match	Ancestry	178393	1183			*Lew	*****@gmail.com
100.9	28.8 Q	3.58	0	0	Match	23andMe	74126	1285	H11A	R-CTS241	*RP Means	*****@aol.com
97.1	28.7 Q	3.60	7.9	7.9	Match	Migration - F2 - T	311740	2319			*Nancy	*****@live.com
93.1	23.8 Q	3.63	0	0	Match	23andMe	141013	1910	H11a	J-M319	Anthony S Evangelista	*****@aol.com

(B)

Name	Ancestral Surnames	Relationship Range	Shared DNA	Longest Block	X Match	Y Haplog	mt Haplog
[redacted]	–	2nd Cousin - 4th Cousin Link on Family Tree	126	49	33	Q-M242	–
[redacted]	[redacted]	2nd Cousin - 4th Cousin Link on Family Tree	100	54	0	R-M269	–
[redacted]	[redacted]	2nd Cousin - 4th Cousin Link on Family Tree	67	58	0	–	–
[redacted]	[redacted]	2nd Cousin - 4th Cousin Link on Family Tree	61	52	0	I-M253	–
[redacted]	–	2nd Cousin - 4th Cousin Link on Family Tree	59	38	0	–	–

FIGURE 7 Example of matches on (A) GEDmatch and (B) FTDNA for different individuals. Both lists show the total amount of shared autosomal and X-chromosome DNA, the length of the longest segment, and any haplogroup information. Clicking on a match's name shows their email address.

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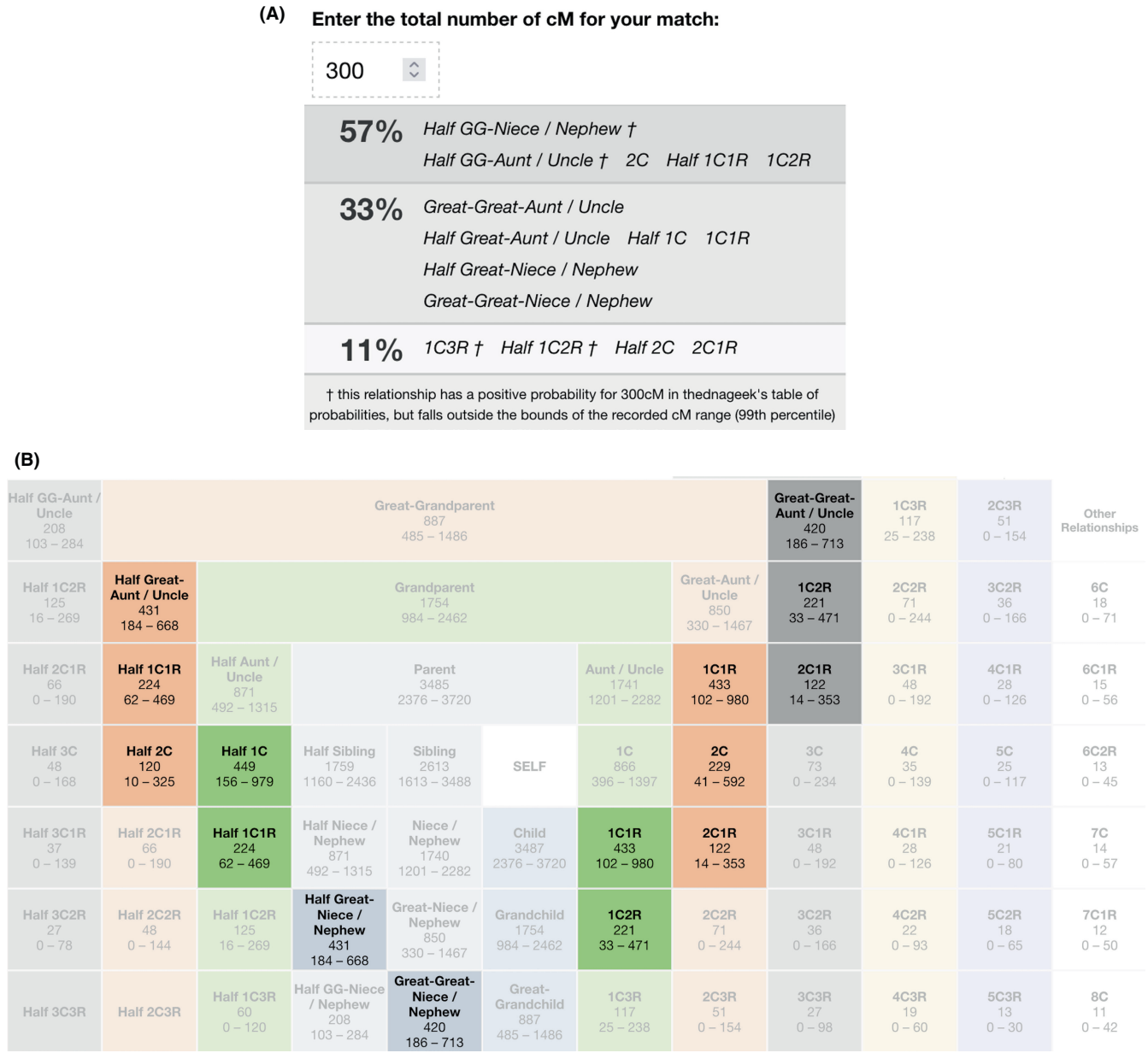


FIGURE 8 Translating shared DNA amounts into relationships using DNA Painter. (A) Relationship probabilities for 300 cM of shared DNA and (B) relationships consistent with that amount of shared DNA (https://dnapainter.com/tools/sharedcmv4/300). Relationships are defined with respect to the "SELF" individual, such that C, cousin and R, removed (e.g., 1C1R, first cousin once removed). Each relationship is shown with the average and range of amounts of shared DNA in cM.

address, can make identification very difficult. A family tree is then built for each of the significant matches in the database back to at least the likely generation of the common ancestor(s) using publicly available information, such as newspaper archives, obituaries, birth records, census records, and social media. Importantly, not all records are accessible, and not every biological relationship is reflected in records; adoption, unknown parentage, and misattributed paternity can obscure a person's biological family and genetic ancestors. Many individuals who test their DNA do so because they have their own family mysteries, which must be solved by the genealogist if that match is to be used to make progress on the case. It is not uncommon to encounter significant roadblocks utilizing one or more of the top matches in the analysis due to genealogical challenges, such as adoptions or misattributed paternities.

Given the 300 cM match in Figure 7, the most likely degree of relatedness is fifth degree (second cousin level, 2C) but could also be fourth degree (first cousin once-removed level, 1C1R) or sixth degree (second cousin once-removed level, 2C1R). Note that each degree of relatedness includes many possible relationships (e.g., a sixth-degree relationship could also be half second cousins (half 2C), half first cousins twice-removed (half 1C2R), or first cousins thrice-removed (1C3R)). Second cousins share great-grandparents, so the genetic genealogist needs to start by building back to that generation. Each individual has four pairs of great-grandparents, and if they

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are second cousins, then the Subject is likely to descend from one set of those ancestors.

Once possible common ancestors have been identified, the genetic genealogist performs descendancy research, in which other descendants of those ancestors are identified. This tree-building and descendancy research analysis is not done for each match in isolation; rather, the database matches are compared with one another to divide them into genetic networks based on which matches share DNA with each other. On GEDmatch, the amount of DNA shared between each pair of matches can be visualized using the matrix tool (Figure 9), while the In Common With (ICW) tool on FTDNA shows whether the matches share *any* DNA. If two or more matches share DNA with one another, then the ancestral line on which they are related is likely the same line that is shared with the Subject (except when endogamy is present in the population group, which means there has historically been a higher than average degree of inter-marrying, resulting in a high level of background relatedness and individuals being related to one another through multiple, more distant relationships instead of a single, closer relationship; see Figure 10 for an example).

In Figure 9, several of the matches share DNA with one another, forming a genetic network. The remaining matches do not share DNA with one another. Somewhere in their family trees, ancestors of all of those matches must intersect via marriages that result in the birth of a child (which are usually reflected in records) or out-of-wedlock births (which typically do not appear in records). These intersections are known as "triangulations" (also called "union couples"), and those triangulations will lead to the Subject. Therefore, when researching descendants, the genetic genealogist also researches the descendants' spouses, looking for connections to other genetic networks and/or single matches that create a triangulation. By painstakingly reconstructing the family trees of the matches, finding triangulations, and searching for descendants of those intersections, the genetic genealogist can narrow down the possible identities of the Subject.

Demographics can assist with narrowing down where the Subject might appear in the matches' trees. The Subject's sex and approximate age are known from examination of the UHR, which limits which generation they could be in. If there is a branch of the family that moved near the area where the UHR was found, that could serve as a clue. Meanwhile, ancestry proportions determined from DNA analysis can indicate the combination of ancestries that must have come together at some point in the family tree, while DNA phenotyping analysis can predict eye color, hair color, skin color, and freckling, which can also narrow down who the Subject could be.

When matches are very distant, it can be extremely challenging to determine which branch of the family tree to focus on. In those cases, it can be valuable to reach out to family members for voluntary, consented samples and/or data uploads to try to zero in on the correct branch. This is variably known as target testing, kinship testing, or reference testing. Family members can also choose to provide family tree information.

2.2.5 | Identity confirmation

In successful cases, the genetic genealogist produces a list of one or more candidate identities for the Subject for the investigating agency

	247.4	148.2	91.6	97	86.8	70.8	72	52.4	45.6	45.3	36.3	41.5
247.4		1080.2	1876.2		3553.6		435.4				1903.9	15.6
148.2	1080.2		507.5		676.1		313.5				304	
91.6	1876.2	507.5			919.8		255.6				563.4	
97												7.6
86.8	3553.6	676.1	919.8				250.7				3546.8	15.9
70.8												7.6
72	435.4	313.5	255.6		250.7						78.1	
52.4												
45.6												
45.3												
36.3	1903.9	304	563.4		3546.8		78.1					
41.5	15.6			7.6	15.9	7.6						

FIGURE 9 GEDmatch matrix of the five GEDmatch matches from Figure 7, along with some more distant matches, showing how much DNA those matches share with one another in cM. Note that the amounts of shared DNA are not exactly the same as in Figure 7, because the matrix tool reports the one-to-one amount of shared DNA, which can be slightly different from the one-to-many amount due to differences in the way they are calculated.

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to follow up on using traditional investigative techniques. Once a likely individual has been identified, DNA testing is used to confirm the identity. When possible, this is done using a traditional STR profile. In the event that a sample from the missing individual or a first-degree relative is not available for STR comparison, the family trees built by the genetic genealogist can be used to identify living close family members who can consent to being tested using SNPs. The much larger number of SNP loci allows for high-confidence kinship testing to more distant relatives than has typically been possible in DVI scenarios, where parents and children are the most highly recommended references, followed only by multiple, more genetically distant relatives such as siblings [20]. The confirmatory test can then be performed locally or using GEDmatch to calculate the amount of shared DNA between the relative and the UHR sample. In dozens of such cases, LE and medical examiners have accepted the results of such a kinship analysis as evidence to support a confirmation of the Subject's identity.

3 | CASE STUDIES

3.1 | 1997 Marshall County, AL John Doe: Jeffrey Kimzey

On April 15, 1997, the mutilated body of a white male was discovered by a fisherman in rural Marshall County, Alabama. The victim was estimated to have been 19–34 years old, 5′9″, and approximately 120 lbs. DNA was extracted from a femur by the Alabama Department of Forensic Sciences, and in December 2019, 3.5 ng was sent to Parabon for WGS. In QC, 83.09% of the sequence aligned to the human genome, so WGE was not necessary and therefore was not performed. Full WGS yielded 12.32X coverage of the human genome, producing 97.79% of the SNPs needed for IGG after imputation. The DNA data was uploaded to GEDmatch returning a top match of 54 cM, which is consistent with a wide range of possible relationships, from second cousin once removed to very distant cousin. Over time, additional DNA matches were connected to the Subject through individuals new to the GEDmatch database and later when the Subject's profile was uploaded to the FamilyTreeDNA database.

As is common in isolated, rural communities, a characteristic of this population is "pedigree collapse." This means that there was a relatively small founding population, with limited marriages occurring outside of this population group over multiple generations. This results in individuals from this population group sharing more DNA in common than is typical. When working in populations with pedigree collapse, it is common to discover multiple relationships between the matches themselves as well as to the Subject. Pedigree collapse can be visualized using the matrix of the matches, which shows small amounts of DNA shared among a large number of the matching individuals (Figure 10).

Despite the challenges of lower matches and the pedigree collapse that occurred within the population group from which the victim descended, trees were constructed for each of the available DNA matches. The surname Burks appeared repeatedly during the course of research, ultimately identifying the likely common ancestor of three DNA matches as William Burks (b. 1743) (Figure 11).

During Parabon's weekly monitoring of the IGG databases, a new match was identified in the FTDNA database that shared 195 cM, which is most likely a second cousin once removed. The match had

	63.4	58	57.4	52.1	47.2	36.6	43	35.1	40	41.3
63.4		39.6	30.4	43.8	7.1	36.9			43.9	33.3
58	39.6		24.1	39.9	7.4	32.6			34.8	15.5
57.4	30.4	24.1		38	20.1	16.6	11.5	11.1	23.8	
52.1	43.8	39.9	38			36.3			39.3	18.7
47.2	7.1	7.4	20.1			24.3	34.9	8.7		
36.6	36.9	32.6	16.6	36.3	24.3				2012.5	21.3
43			11.5		34.9					
35.1			11.1		8.7					
40	43.9	34.8	23.8	39.3		2012.5				19.1
41.3	33.3	15.5		18.7		21.3			19.1	

FIGURE 10 GEDmatch matrix of the Marshall County, AL John Doe's top matches showing small amounts of DNA shared among many of the individuals (measured in cM), indicating pedigree collapse and complicating the IGG research.

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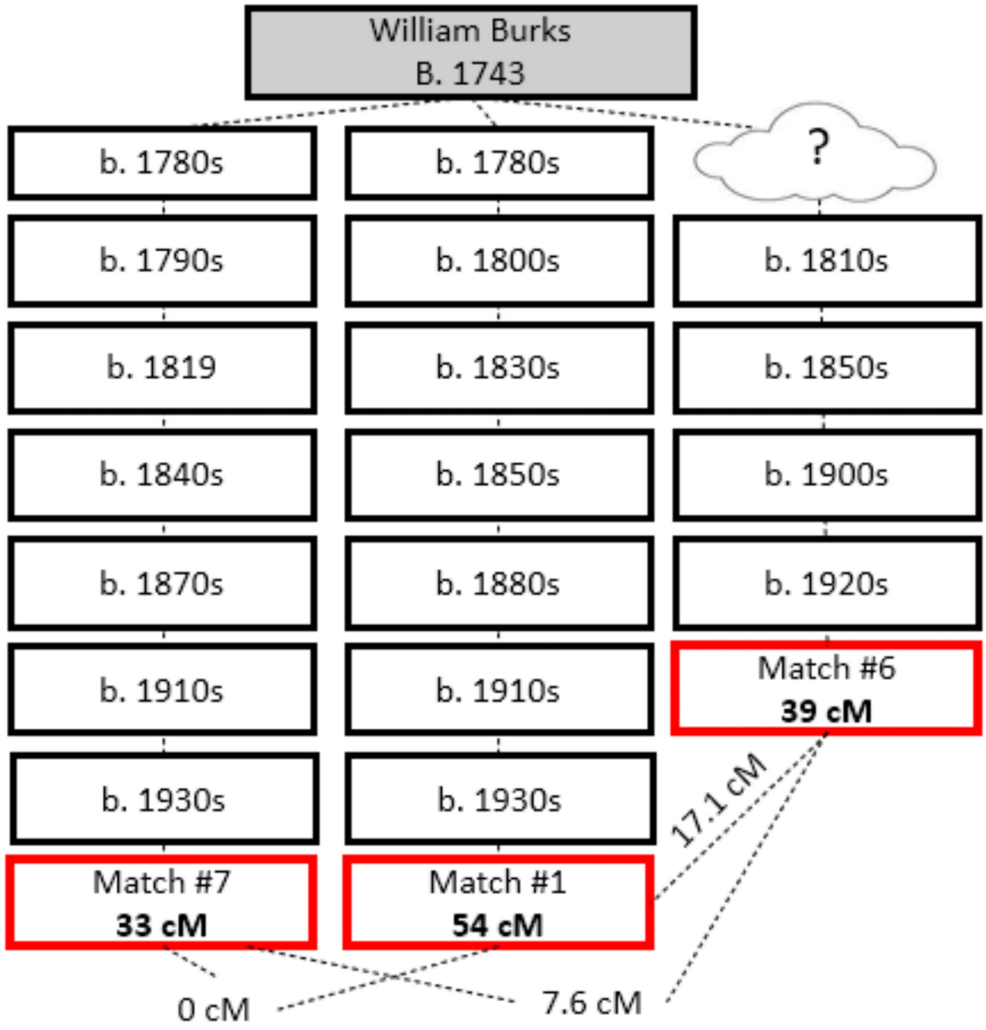


FIGURE 11 Family tree of the original top matches of the Marshall County, AL John Doe, showing their descent from a common ancestor, William Burks (gray).

an untraceable email address, and significant research went into identifying the correct individual as there were more than a dozen individuals in the United States with the same name. After the agency requested additional support from another state concerning vital records, the identity of the match was confirmed. Parabon's genetic genealogist set to work constructing a detailed tree back four generations and then performed descendancy research looking for any connections to other matches and/or the Burks family.

While sometimes uncommon surnames can be a red herring, in this case, an uncommon surname led to the identification. While researching the family tree of the new match, a distant cousin was connected to the new match's ancestor, David Kimzey (b. 1755) (Figure 12), whose surname was subsequently identified in another distant match's family tree. Based on the genetic distance between this new match and the Subject, the descendants of Match #1's great-grandparents, Lewis Kimzey (b. 1896) and Edna Jones (b. 1897), were exhaustively researched. One son of the couple, Edward Kimzey (b. 1916), married Gladys Estalee Sweet (b. 1921), daughter of Luther Sweet (b. 1902) and Cora Burks (b. 1901). The surname Burks was immediately recognized and it was discovered that Cora was the daughter of William Benton Burks (b. 1866) and Laura Ellen Burks (b. 1872). This couple were double second cousins twice-removed to each other through their shared ancestor David Burks, the son of William Burks (b. 1743), the common ancestor of genetic network #1. Whereas second cousins share a single pair of great-grandparents, double second cousins share two of their eight pairs of great-grandparents. This key triangulation between the genetic networks means that the unidentified individual was highly likely to descend from the marriage of Edward Kimzey and Gladys Sweet (Figure 13).

Since the couple had only two sons and both were alive after the remains of the Marshall County, AL John Doe were discovered, a grandson of the couple was more likely to be the Subject. Only one grandson appeared to be missing, Jeffrey Kimzey (b. 1977) of San Marcos, California, who would have been 20 years old when the remains were discovered. Additional research into the life of Jeffrey Kimzey was performed and no additional public records were found for him after 1996. When his grandmother died in 2011, he was listed as a survivor, but his name did not include his location, unlike the other survivors listed. It seemed evident that the family

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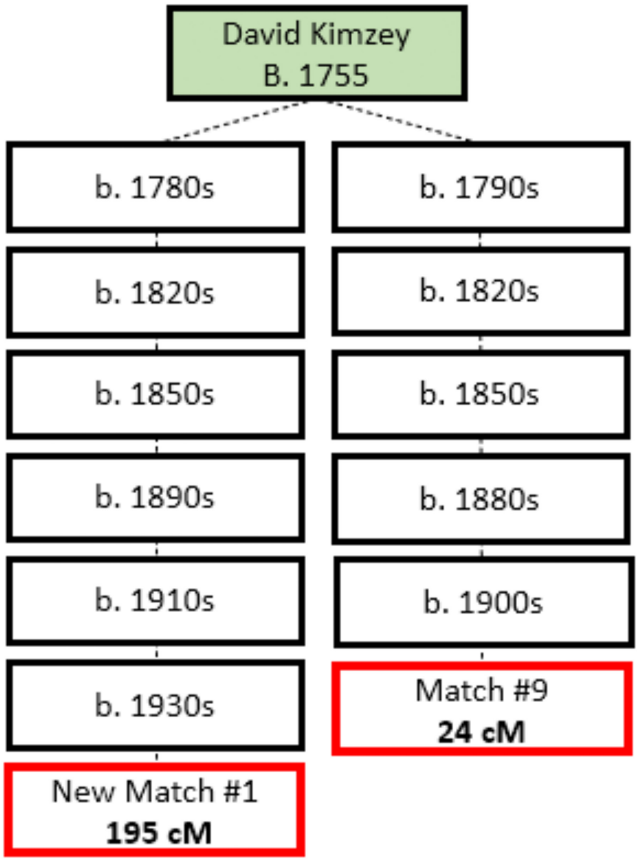


FIGURE 12 Family tree of a new cousin (Match #1) and a distant cousin (Match #9) of the Marshall County, AL John Doe, showing their descent from a common ancestor, David Kimzey (green).

did not know his whereabouts. A yearbook search was performed and a 1996 yearbook photograph was discovered that showed Jeffrey Kimzey at San Marcos High School in Southern California (Figure 14). The Marshall County Sheriff's Office, with assistance from the Alabama Department of Forensic Sciences, confirmed the remains belonged to Jeffrey Kimzey using DNA reference samples from each of his parents. Jeffrey's death has been deemed a homicide and the investigation into his murder continues.

3.2 | 1979 Multnomah Falls, OR John Doe: Freeman Isaac Asher, Jr

On September 9, 1979, hikers discovered the skeletal remains of an African American male on Climber's Descent Trail near Multnomah Falls east of Portland, Oregon. The remains revealed little about this unidentified man in 1979; he was estimated to be 70″ tall and was between 20 and 35 years old. Investigators believed he may have been deceased for over a year. His death did not appear suspicious; it was theorized that he may have fallen while hiking a less-traveled trail. His hat, a yellow ball cap with black felt letters NT, was the most distinctive item found with his body, and investigators initially attempted to identify him through the origin of this hat, without success.

In 2020, DNA was extracted from a tooth, and 10 ng was sent to Parabon for DNA processing, resulting in a 79.3% call rate. Ancestry inference indicated that the individual was of African American descent. The DNA data were uploaded to GEDmatch,

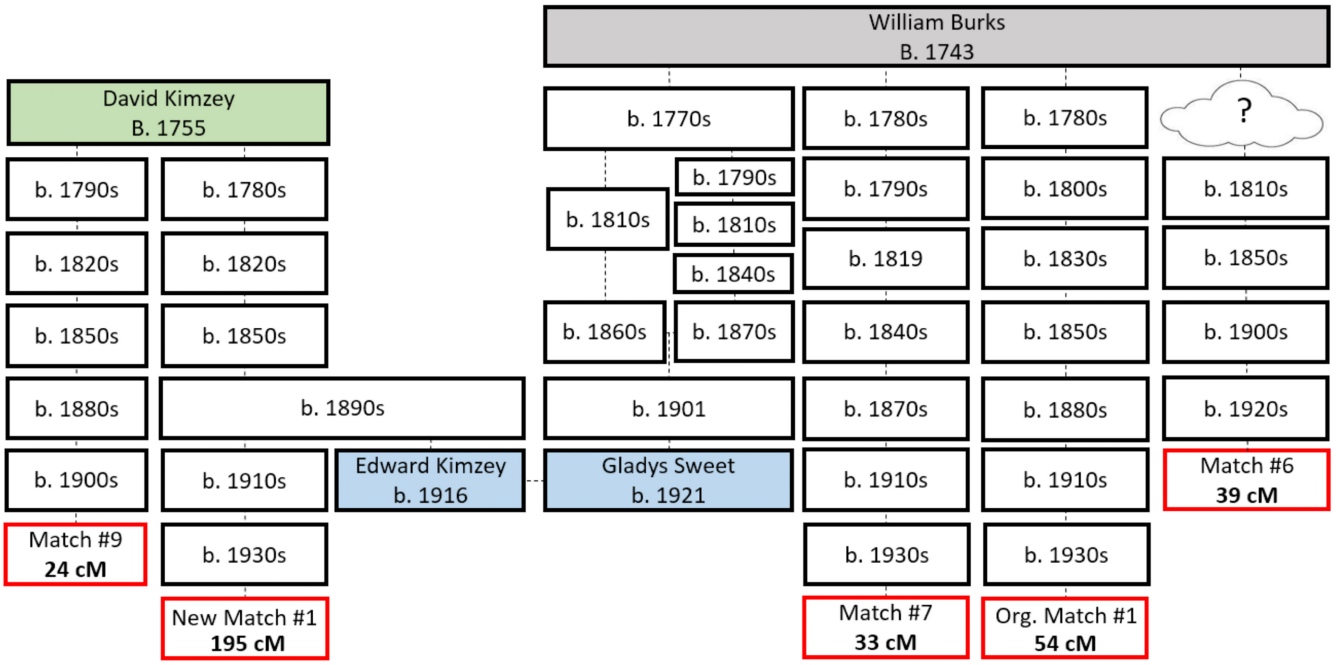


FIGURE 13 Triangulation of the Marshall County, Alabama John Doe's top matches occurs at the marriage of Edward Kimzey and Gladys Sweet (blue), descendants of the common ancestors David Kimzey (green) and William Burks (gray).

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where the top match shared 114 cM of autosomal DNA. This top GEDmatch match was most likely a second cousin once removed or third cousin. Considering the age of the case, this meant that their common ancestor was likely born prior to the end of slavery, which could seriously complicate the research. Additionally, if the Subject was 25 in 1979, he was born after the 1940 US Federal Census, which was the last census that was available for genealogy research at that time. The US Census is released 72 years after it was taken, so the 1950 census is now available, but it was not at the time research was performed.

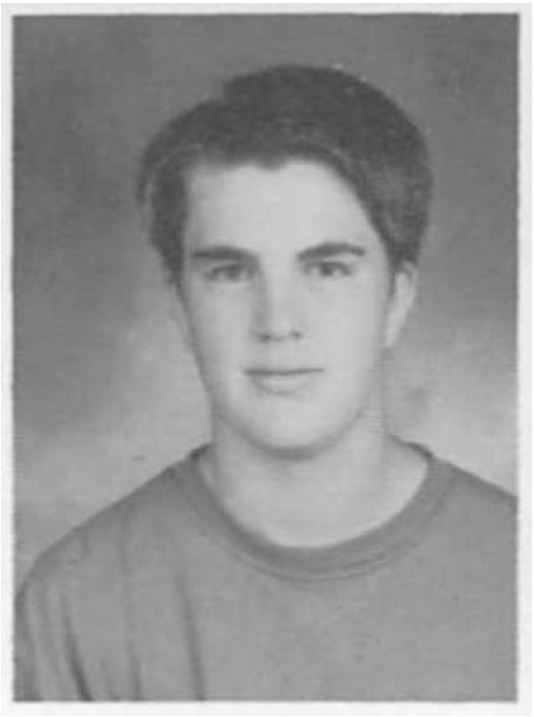


FIGURE 14 A 1995 San Marcos High School yearbook photograph of Jeffrey Douglas Kimzey, the Marshall County John Doe.

While researching the original top match, a new match sharing 303 cM uploaded her DNA data to the GEDmatch database. This new match was a much closer cousin to John Doe and could be the key to unlocking his mystery. She was likely to be a second cousin (or genetic equivalent), but unfortunately, this new match had questions about her own heritage, as is common among people who participate in genetic genealogy. With significant research, the analyst found that the new match's maternal grandfather was related to the original top match.

Building the family trees of the matches led to a common ancestral couple, William Spencer Harrison (b. 1832) and Moriah Simmons (b. 1842), which means the Multnomah Falls John Doe is highly likely to also descend from this couple (Figure 15). Descendancy research was performed in an attempt to identify all male descendants of this couple that were born between 1940 and 1959; Parabon's genetic genealogist discovered that Freeman Isaac Asher, Jr (b. March 15, 1944) was a great-grandchild of the common ancestral couple and a first cousin twice-removed to the top match. There were no references found to him in public or social records after 1971. He had not been reported as a missing person to any agency.

With guidance from Parabon's IGG team, the agency reached out to a great-nephew of Freeman Asher who appeared to be the family genealogist. The family confirmed that Freeman Isaac Asher, Jr. (Figure 16) was missing, and the last memory anyone has of

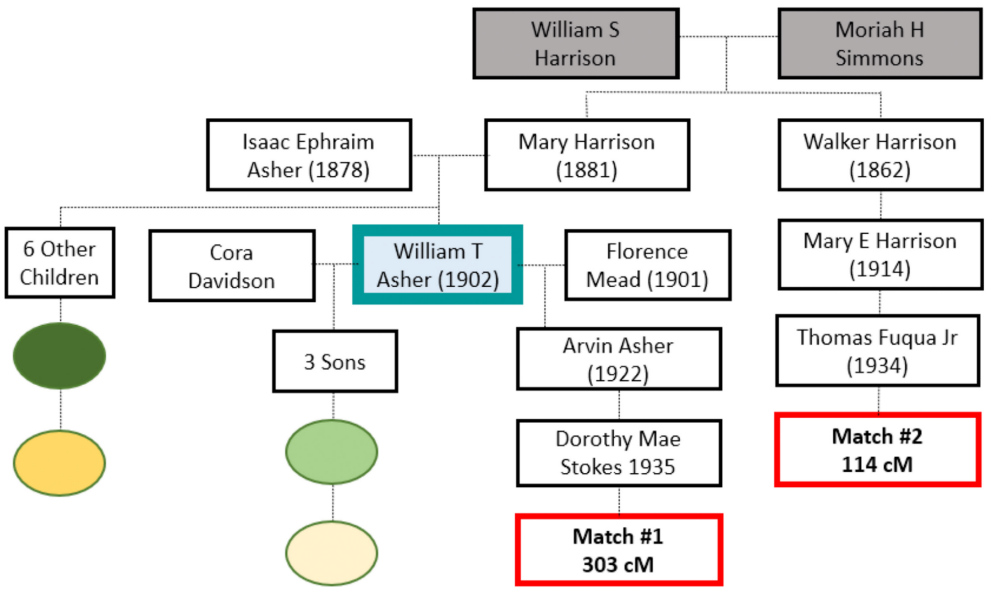


FIGURE 15 Family tree of the Multnomah Falls, Oregon John Doe's top matches, showing their common descent from an ancestral couple (gray) and the most likely (green) and less likely (yellow) positions the John Doe could occupy in the tree. Based on the amount of DNA shared with Match #1, the Subject could be the grandson or great-grandson of William Asher (blue) and his other partner, Cora Davidson, or descend from one of William's siblings.

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Freeman was him singing 'Easy' by the Commodores in March 1977. Freeman's surviving full sibling provided a DNA sample, which was sent for microarray genotyping and resulted in 2730 cM of shared atDNA and 43.1 cM shared on the X-chromosome. This result is exclusively consistent with a full sibling relationship, which was used to support the Medical Examiner's identification of the Multnomah Falls John Doe as Freeman Isaac Asher, Jr. Subsequently, the agency released Freeman's remains to his surviving family.

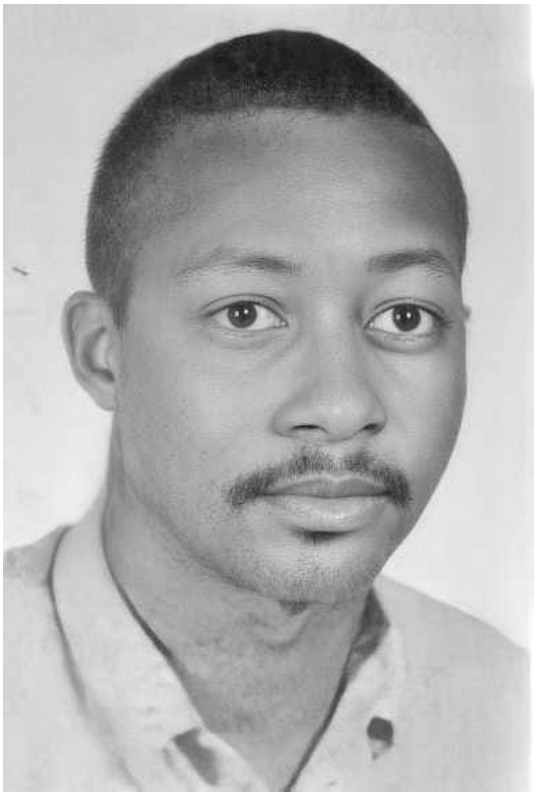


FIGURE 16 Yearbook photograph of Freeman Isaac Asher, Jr, the Multnomah Falls John Doe.

3.3 | 1988 Santa Cruz County, CA John Doe: Marty Rupar

On Saturday, February 13, 1988, trash bags containing the partial skeletal remains of a young adult male were found in a ravine off Bonny Doon Road in Santa Cruz County, California. With no leads as to the identity of the remains, the Santa Cruz County Sheriff's Office labeled the remains as a John Doe.

In June 2020, DNA was extracted from a tooth belonging to this John Doe and 14 ng was sent to Parabon for WGS with WGE, which yielded 4.1X coverage of the human genome and allowed 92.9% of the targeted SNPs for IGG to be called after imputation. The profile was uploaded to GEDmatch, where the top match shared 117 cM of atDNA—most likely a second cousin once removed to the third cousin.

The genetic genealogist constructed trees for each of the significant DNA matches in GEDmatch, leading to the conclusion that John Doe descended from two sets of common ancestors. The first set of common ancestors identified was Merretta Quimby (b. 1848) and John Close (b. 1853) (Figure 17). The Subject was likely Merretta and John's great-great-grandson, and at this level, there were many descendants to identify. In Merretta's obituary, her and her husband's survivors included nine children, 36 grandchildren, 150 great-grandchildren, and *300 great-great-grandchildren*!

The other common ancestral couple was identified as Cisaly Manion (b. 1798) and Hugh Dycus (b. 1793), albeit with far fewer genetic connections to provide guidance. An additional DNA match further indicated that the Subject was very likely to descend from a specific grandson of the common Dycus ancestors, Isaac Dycus (b. 1877) and his wife, Flora Anderson (b. 1878) (Figure 18).

In an effort to identify a triangulation between the descendants of the two genetic networks, genealogical research was conducted for each spousal line added during descendancy research. Unfortunately, no connection between the families of Merretta and John Close of Wisconsin and Flora Anderson and Isaac Dycus

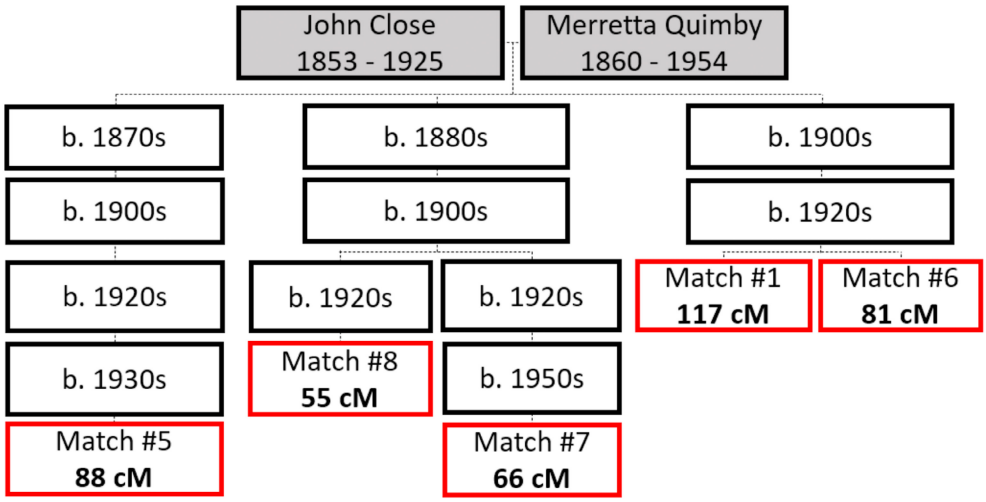


FIGURE 17 Family tree of five of the Santa Cruz County, CA John Doe's top matches (red boxes), illustrating their descent from a common ancestral couple (gray).

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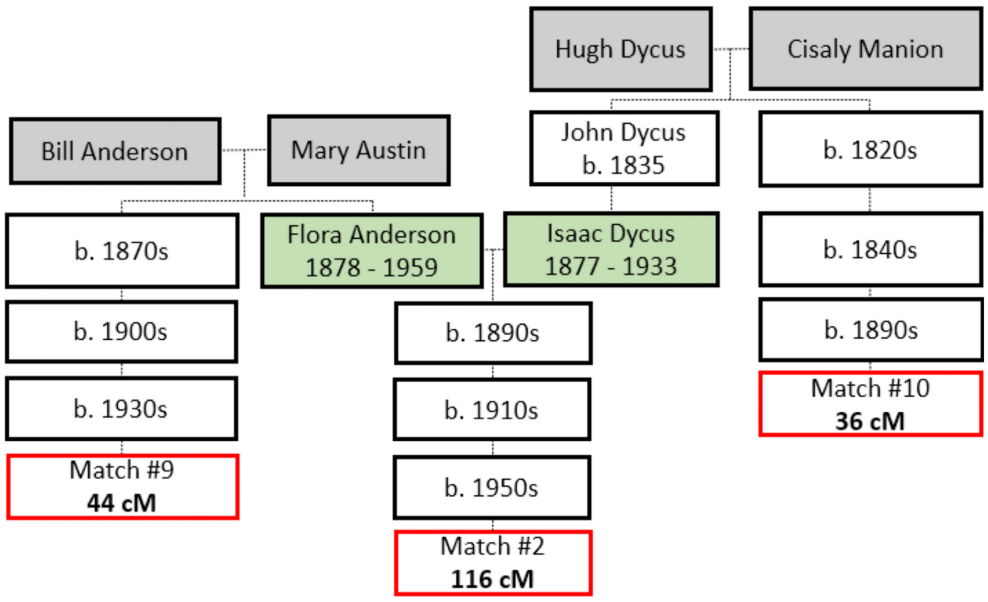


FIGURE 18 Family tree of three of the Santa Cruz County, CA John Doe's top matches (red boxes) and their convergence at the marriage of Flora Anderson and Isaac Dycus (green).

of Tennessee was discovered. Individuals that were related to each of the families were contacted and requested to assist with a LE investigation of a John Doe by uploading their DNA to GEDmatch in comparison with the Subject's genetic file. These voluntary uploads resulted in closer matches, and led the investigation to focus on the neighborhoods of residence and the employment of each of the descendants of the two ancestral couples. This research led to Doris Mae Hall, who was the granddaughter of Nellie Close (b. 1879) and the great-granddaughter of Merretta and John Close. Doris was a bartender from Green Bay, WI and, according to newspaper reports, Doris worked at the "Green Parrot Bar". Coincidentally, the grandson of Isaac Dycus (b. 1877), John William Dycus, was the owner of that bar. While there were no documented children of this couple in Wisconsin birth records, the geographic proximity and shared place of employment of a descendent of each of the two identified ancestral couples was a significant clue that the Subject could be a son born from a relationship between them.

Parabon's genetic genealogist scoured public records for any potential missing person connected to Doris Hall. Doris's children were named in her obituary, including her son, Marty Robin Rupar (b. May 1966) (Figure 19), who, although his father was listed in records as being Doris' husband, appeared to be a possible candidate to be the Subject. With the assistance of the Santa Cruz County Sheriff's Office, it was determined that Marty Rupar had not been seen since the 1980s. The family confirmed Parabon's theory that John Dycus was actually the biological father of Marty Rupar.

To confirm the hypothesis developed through IGG, the agency sought DNA from living relatives of Marty Rupar, and several new matches uploaded their DNA. A DNA sample from Marty's maternal half-sibling was found to share 1803 cM of atDNA and 97.7 cM on the X-chromosome, which is consistent with the theorized relationship and was accepted by the sheriff's office as evidence to identify



FIGURE 19 Clay model of the Santa Cruz County John Doe created by the National Center for Missing and Exploited Children (NCMEC), with pigmentation predictions provided by the Snapshot® DNA Phenotyping system (left). A photograph of Marty Rupar was obtained by investigators after Parabon's lead was provided (right).

the Bonny Doon Road John Doe as Marty Rupar and close the case (Figure 20). Marty's family members expressed their appreciation for the opportunity to lay him to rest. Investigators are still actively investigating the circumstances of Marty Rupar's death.

4 | CONCLUSION

Investigative genetic genealogy has proven to be an extraordinary tool for the identification of unidentified human remains. While it can be challenging to obtain genome-wide SNPs from UHR DNA,

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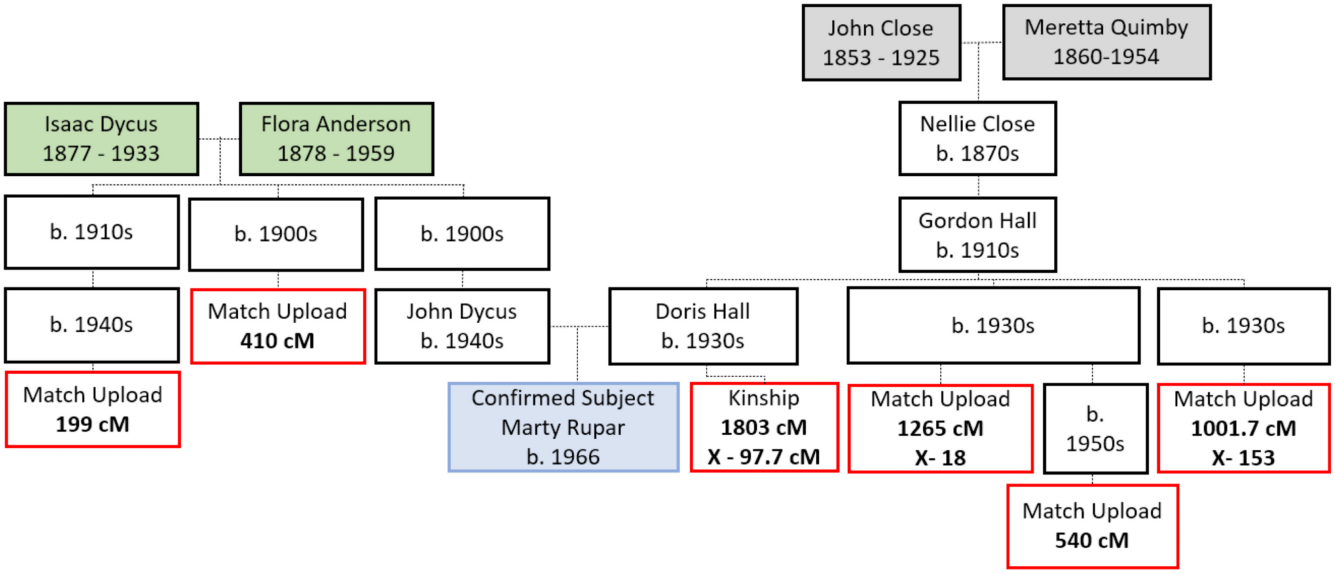


FIGURE 20 Key triangulation of the DNA matches provides confirmation that the Santa Cruz County, CA John Doe is the son of John Dycus and Doris Hall, Marty Rupar (b. 1966).

which is often highly degraded and contaminated, new laboratory and bioinformatic techniques, such as human genome enrichment and imputation, have led to hundreds of UHR identifications through the use of IGG since 2018. These same proven techniques, which have been developed and refined for the challenging DNA samples encountered in UHR cases, have great potential to be used for DVI, particularly when the DNA is compromised and/or family reference samples are not available.

ACKNOWLEDGMENTS

The authors would like to thank the Marshall County, AL Sheriff's Office, the Oregon State Medical Examiner's Office, and the Santa Cruz County, CA Sheriff's Office for allowing us to use their cases as case studies.

CONFLICT OF INTEREST STATEMENT

All authors are employees of Parabon NanoLabs, which provides DNA processing, bioinformatics, and investigative genetic genealogy services to law enforcement.

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How to cite this article: Greytak E, Wyatt S, Cady J, Moore C, Armentrout S. Investigative genetic genealogy for human remains identification. J Forensic Sci. 2024;69:1531–45. https://doi.org/10.1111/1556-4029.15469

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